

Ballarat High School



Data Analysis – Application Task (2022)

Part 2: The Competition

This Data Analysis Application Task is designed to run in 2 parts:

Part 1: Matilda's Bakery (100 Minutes - 5 Minutes Reading and 95 Minutes Writing)

Part 2: The Competition (100 Minutes - 5 Minutes Reading and 95 Minutes Writing)

Note:

All answers must be written in the answer booklet provided.

Matilda's Bakery



Outcome 1

Define and explain key concepts and apply related mathematical techniques and models as specified in Area of Study 1 in routine contexts. (10 marks)

Outcome 2

Select and apply the mathematical concepts, models and techniques as specified in Area of Study 1 in a range of contexts of increasing complexity. (20 marks)

Outcome 3

Select and appropriately use numerical, graphical, symbolic and statistical functionalities of technology to develop mathematical ideas, produce results and carry out analysis in situations requiring problem-solving, modelling or investigative techniques or approaches. (10 marks)

Section A

For this section, we are looking at **Age vs. Number of Competitions Competed In** (Table 1).

- a) A grid has been provided with the axes labels and scale completed. Complete the scatterplot by adding the relevant data from the table. (2)
- b) Which of the variables being investigated is the explanatory variable? (1)
- c) Comment on the strength and direction of this relationship. (1)
- d) Calculate and write the least squares line of best fit equation to two significant figures, using the variables under investigation. (2)
- e) Plot the calculated Least Squares line on the display created. (1)
- f) Interpret the **y-intercept** and **slope** for the calculated least squares line. (4)
- g) The judges are considering adding two new competitors to the line-up. Martha is 42 years old, and Ricciardo is 15. How many competitions does the model predict each competitor will have competed in? Round each answer to the nearest whole number. (2)
- h) Which of the predictions from **Part g** is more reliable? Justify your answer. (2)
- i) Showing all working, find the residual value for Guntram (Round answer to 2 decimal places) (2)

Section B

For this section, we are looking at the relationship between **1st Round Points** and **2nd Round Points** (Graph 1). A completed scatterplot and r^2 value have been given to you in the data sheet.

- a) Using the information provided, calculate the Pearson Correlation Coefficient. (Round answer to 2 decimal places) (1)
- b) What % of the variation in the 2nd Round Points is **not** explained by the variation in 1st Round points? (Round to nearest percentage) (1)
- c) Using the line of best fit on the graph, estimate the equation for line of best fit for this data, using the variables under investigation. Round your coefficients for slope and y-intercept to two significant figures (2)

Section C

For this section, we are looking at the effect that **Years of Baking Experience** has on **1st Round Points** won (Table 1).

- a) Using the residual plot, comment on the assumption of linearity for this data set justifying with appropriate evidence. (2)
- b) We want to linearise this data set. List all possible transformations that could be tested for this shape graph? (1)
- c) A squared transformation is applied to the variable, 1st Round Points. Apply this squared transformation to the data and determine the equation of the least squares regression line that allows 1st Round Points to be Predicted from Years of Experience. Round coefficient values to two significant figures. (2)
- d) Use this equation to predict the Number of 1st Round Points for a competitor with 26 Years of Experience. (Round to nearest whole number) (1)
- e) Has this squared transformation created a more reliable model to predict with? State your contention using a piece of evidence to justify why/why not. (1)

Section D

For this section, we are looking at the trend over time for the **Number of Competition Entries per Year** (*Table 2*), and the **Daily Revenue Trends** of our favourite competitor, Matilda (*Table 3*).

- a) Construct a time series plot on the grid provided showing the Number of Competition entries over the last 15 years using coded time as the explanatory variable. The scale and label for response variable has been given. (3)
- b) Irregular fluctuations are present within the time series data. List two other features present within the time series plot. (1)
- c) Smooth the data using **3-median** smoothing on the plot created. (2)
- d) Show all working for the **2-mean with centring** smoothed value of 2013. (1)
- e) Based on the graph, would you suggest median smoothing or mean smoothing to smooth the data set? Justify your answer (1)

Matilda wishes to test her seasonal index knowledge on her Daily Revenue shown in *Table 3*.

- f) Find the missing seasonal index for Friday. (1)
- g) Interpret the seasonal index for Sunday in terms of the variables under investigation. (1)
- h) To correct for seasonality, what percentage (to the nearest %) should the daily revenue for Wednesday be increased? (1)
- i) Matilda predicts the deseasonalised daily revenue for Wednesday of Week 4 to be \$440. Find the actual daily revenue (to the nearest dollar) for Wednesday based on this deseasonalised prediction. (1)